π_RL · arXiv 2510.25889 · Oct 5

Scientific Reviewer

8+3 · seat weight 3 · HIGH

7 actionable critiques as THE talk figure
Strong ID mastery · partial env-shift

Kanta not seated — all five packs built

Spine: unlock PPO on flow VLAs; BEHAVIOR still prefers RFT

Chen / Liu / Zhang et al. · RLinf · ICML-style (TENTATIVE)

Club spine · Oct 5

Unlock PPO on flow VLAs · BEHAVIOR still prefers RFT

BLOCKED

Flow VLAs

intractable logπ
+ deterministic ODE
→ no PG / no explore

→

π_RL OPENS

Flow-Noise / SDE

tractable Gaussian
+ PPO/GAE on ~300M
expert (VLM frozen)

→

FAST SIM

LIBERO / ManiSkill

+27–31 pp ID avg
few-shot+RL > full SFT
partial env OOD

→

SLOW SIM

BEHAVIOR / OmniG

Comet cites π_RL
then uses RFT
sample-efficiency gate

When to use π_RL

- Parallel sparse-success sim available
- OpenPI π_0 / $\pi_{0.5}$ (or GR00T) already SFT'd
- Want ID specialists beyond demos
- Can afford 8×H100-class + 64–320 envs
- Real deploy = sim RL → ODE eval (not real RL)

When to stay on RFT

- Slow household sims (BEHAVIOR)
- Heterogeneous GPU / RT-core split pain
- Need coverage flywheel, not on-policy
- Comet path: reject→SFT (Q 0.224→0.345)
- 2026: compose RFT then short π_RL burst

Strengths · Scientific Reviewer

Real gap

Flow VLAs blocked policy gradients — correctly posed

Two mechanisms

Flow-Noise + Flow-SDE with clear ancestry + tradeoffs

Broad empirics

4 suites + OOD + SIMPLER + real + GR00T + HP grids

Honest OOD

Visual/env works; novel tasks (ML45) fail — no overclaim

Open code

RLinf + HF + unusually complete App J hypers

Verdict preview

Strong same-task mastery + partial env-shift

Don't overread as open-world / cross-task discovery

7 actionable critiques · THE Reviewer talk figure

From KANTA_PACK — speak these, not abstract praise

1 **Env-steps / GPU-hours**

Report steps to plateau vs 'beats full SFT' — interactive cost missing

2 **Unfreeze last VLM layers**

Frozen VLM may cap visual OOD — ablate LoRA on high visual-shift

3 **Noise vs SDE decision rule**

Wins differ by suite — need rule or multi-seed, not cherry averages

4 **Exact likelihood clarity**

Exact for augmented denoise MDP — not marginal ODE policy

5 **Thin real n (40%)**

More seeds / tasks / CI — one pick-place is anecdotal

6 **Head-to-head FPO / RFT**

Matched budgets vs advantage-weighted CFM and Comet-style RFT

7 **Don't overclaim venue**

ICML-style keywords; acceptance TENTATIVE — say so

OOD honesty · what transfers

TRANSFERS

- ManiSkill visual / layout / exec shifts
- CALVIN ABC→D scene transfer
- Partial relative OOD gains (~+100%)
- Preserves SFT OOD skills (less forgetting)

DOES NOT

- MetaWorld ML45 novel tasks — oscillates
- Cross-task skill invention via RL
- Open-world discovery narrative
- Visual OOD ceiling (frozen VLM)

Reading for Reviewer + Empiricist

RL = action-level refinement on seen skills

Not a substitute for multi-task demos or stage structure

BEHAVIOR implication: recovery / temporal efficiency — not 50-task invention

Table 1 · ID average success % · tattoo these

Four suites averaged · Δ vs few-shot SFT

Model	Method	LIBERO	ManiSkill	MetaWorld	CALVIN	Avg	Δ
π_0	SFT	57.6	38.4	50.8	57.5	51.1	—
π_0	Flow-SDE	96.1	78.8	78.1	61.7	78.7	+27.6
π_0	Flow-Noise	97.6	77.8	85.8	59.9	80.3	+29.2
$\pi_{0.5}$	SFT	77.1	40.1	43.8	61.3	55.6	—
$\pi_{0.5}$	Flow-SDE	97.9	90.9	70.7	87.0	86.6	+31.0
$\pi_{0.5}$	Flow-Noise	98.3	89.7	66.1	84.5	84.7	+29.1

π_0 SFT→Noise

51.1 → 80.3 (+29.2)

$\pi_{0.5}$ SFT→SDE

55.6 → 86.6 (+31.0)

Best avg

$\pi_{0.5}$ Flow-SDE 86.6

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https://github.com/RLinf/RLinf https://huggingface.co/RLinf

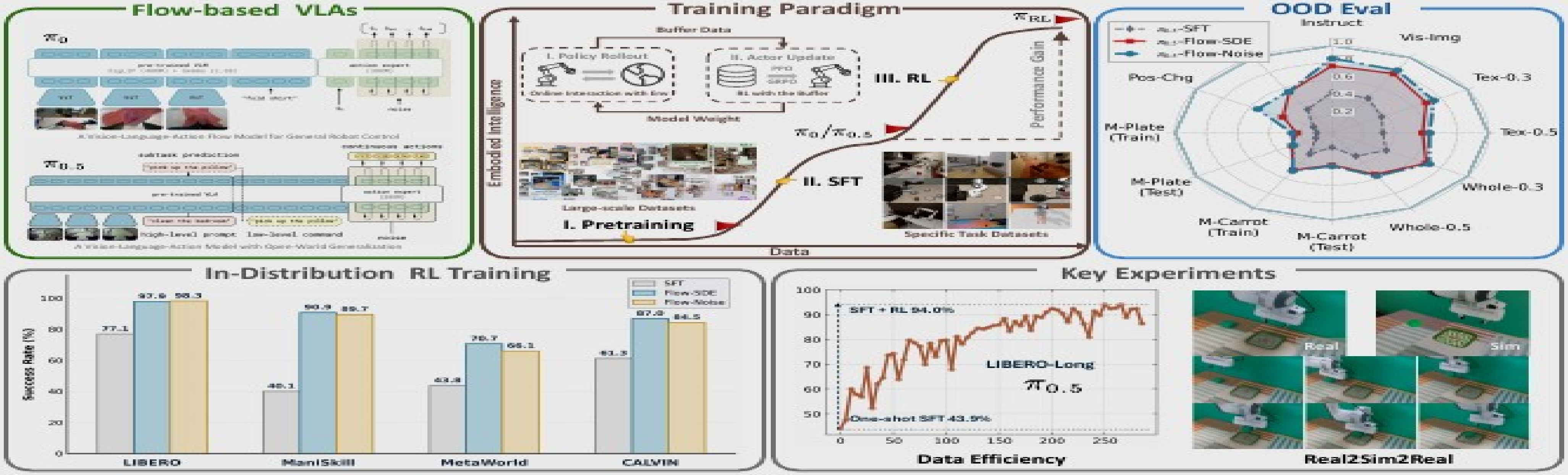

Figure 1. π_{RL}: An online RL framework for flow-based VLAs. Incorporating two solutions, Flow-Noise and Flow-SDE, π_{RL} enhances the performance and generalization of SFT-aligned models across extensive ID benchmarks and OOD settings. Refined with RL, few-shot SFT policies achieve performance comparable to full dataset baselines. Additionally, we facilitate seamless zero-shot sim-to-real transfer by constructing a simulator with 3D Gaussian Splatting as the rendering engine to narrow the visual domain gap.

Abstract

Vision-Language-Action (VLA) models enable robots to understand and perform complex tasks from multimodal input. Although recent work explores using reinforcement learning (RL) to automate the laborious data collection process in scaling supervised fine-tuning (SFT), applying RL to large-scale flow-based VLAs (*e.g.*, π_0, $\pi_{0.5}$) remains challenging due to intractable action log-likelihoods raised from flow matching. We address this challenge with π_{RL}, featuring two technical approaches: (1) **Flow-Noise** models the denoising process as a discrete-time MDP with a learnable noise network for exact log-likelihood computation. (2) **Flow-SDE** integrates denoising with agent-environment interaction, formulating a two-layer MDP that employs ODE-to-SDE conversion for efficient RL exploration. We evaluate π_{RL} across various benchmarks, with experiments demonstrating that RL yields significant performance improvements in both in-distribution and out-of-distribution settings.

[1]Tsinghua University [2]Peking University [3]Institute of Automation, Chinese Academy of Sciences [4]Carnegie Mellon University [5]Infinigence AI [6]Zhongguancun Academy ♯ Work done at Tsinghua University ∗ Equal Contributions . Correspondence to: Yu Wang <yu-wang@tsinghua.edu.cn>, Chao Yu <zoeyuchao@gmail.com>.

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Flow-Noise vs Flow-SDE · two ways to unlock PPO

Both train stochastic · both deploy as deterministic ODE (noise discarded / SDE off)

Flow-Noise

ReinFlow-inspired · one-layer MDP

- Learnable $\sigma\_\theta'(A^\tau, o)$ → Gaussian denoise step
- Exact joint log-prob of denoise chain (eq. 5)
- joint_logprob=True · entropy ~0.005
- Noise net discarded after fine-tune
- Often wins MetaWorld (π_0 85.8 vs 78.1)
- Cost: full denoise recompute for likelihood

Flow-SDE

Flow-GRPO / DPPO · two-layer + hybrid

- ODE → equivalent SDE (Song PF-ODE)
- Noise $a\sqrt{(\tau/(1-\tau))}$; Gaussian step likelihood
- Two-layer MDP: denoise inner + env outer
- Hybrid: 1 SDE step / env · rest ODE → ~2×
- Default efficiency pick (paper preference)
- Often wins CALVIN $\pi_{0.5}$ Len-5 (87.0)

Shared recipe

few-shot SFT → freeze VLM → PPO/GAE on ~300M action expert · chunk reward · RLinf · 8×H100

Control note: chunk length H = credit-assignment knob · not classical planner

Reviewer verdict

Strong same-task mastery

+ partial env-shift robustness

Don't overread as open-world discovery

ACCEPT-LEANING

methods+systems
if sample-efficiency
+ multi-seed improve

WHAT IT SHOWS

ID polish after
few-shot SFT;
env/visual OOD partial

WHAT IT DOESN'T

novel-task invention;
real online RL;
iso-compute vs SFT

Reviewer close

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Unlock PPO on flow VLAs · BEHAVIOR prefers RFT

π_{RL}: Online RL Fine-tuning for Flow-based Vision-Language-Action Models

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